```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df=pd.read_csv("/content/drive/MyDrive/Skillcraft Technology/sample-marketing_origional_dataset.csv")
```

```
print("Dataset Shape:")
print(df.shape)
```

```
Dataset Shape:
(260, 8)
```

```
print("First five rows:")
df.head()
```

First five rows:

| | date | channel | campaign | impressions | clicks | spend | conversions | revenue |
|---|---|---|---|---|---|---|---|---|
| **0** | 2026-02-24 | Organic | Brand-9 | 24549 | 1392 | 1698.24 | 45 | 4632.75 |
| **1** | 2026-01-23 | Organic | Brand-1 | 49123 | 2235 | 4559.40 | 40 | 3373.60 |
| **2** | 2026-03-02 | Email | Spring-4 | 83777 | 1215 | 1518.75 | 105 | 12655.65 |
| **3** | 2026-01-08 | Organic | Retargeting-6 | 9714 | 445 | 1388.40 | 30 | 3683.40 |
| **4** | 2026-02-21 | Email | Brand-6 | 45743 | 2594 | 5473.34 | 154 | 32731.16 |

```
print("Last five rows:")
df.tail()
```

Last five rows:

| | date | channel | campaign | impressions | clicks | spend | conversions | revenue |
|---|---|---|---|---|---|---|---|---|
| **255** | 2026-04-25 | Email | Launch-1 | 66624 | 3957 | 10565.19 | 306 | 6851.34 |
| **256** | 2026-01-17 | TikTok | Launch-6 | 71403 | 1942 | 6350.34 | 97 | 20690.10 |
| **257** | 2026-02-23 | LinkedIn | Retargeting-3 | 66566 | 2443 | 3102.61 | 44 | 5178.36 |
| **258** | 2026-02-24 | TikTok | Retargeting-4 | 5778 | 153 | 512.55 | 5 | 417.70 |
| **259** | 2026-04-30 | LinkedIn | Always-On-5 | 44625 | 2628 | 1787.04 | 161 | 16030.77 |

```
print("Dataset Information")
print(df.info())
```

```
Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 260 entries, 0 to 259
Data columns (total 8 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   date         260 non-null    object
 1   channel      260 non-null    object
 2   campaign     260 non-null    object
 3   impressions  260 non-null    int64
 4   clicks       260 non-null    int64
 5   spend        260 non-null    float64
 6   conversions  260 non-null    int64
 7   revenue      260 non-null    float64
dtypes: float64(2), int64(3), object(3)
memory usage: 16.4+ KB
None
```

```
df['date'] = pd.to_datetime(df['date'])
```

```
print("Dataset Information")
df.info()
```

```
Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 260 entries, 0 to 259
Data columns (total 8 columns):
```

```
 #   Column       Non-Null Count  Dtype         
---  ------       --------------  -----         
 0   date         260 non-null    datetime64[ns]
 1   channel      260 non-null    object        
 2   campaign     260 non-null    object        
 3   impressions  260 non-null    int64         
 4   clicks       260 non-null    int64         
 5   spend        260 non-null    float64       
 6   conversions  260 non-null    int64         
 7   revenue      260 non-null    float64       
dtypes: datetime64[ns](1), float64(2), int64(3), object(2)
memory usage: 16.4+ KB
```

```
print("Column Names:")
print(df.columns.to_list())
```

```
Column Names:
['date', 'channel', 'campaign', 'impressions', 'clicks', 'spend', 'conversions', 'revenue']
```

```
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
date           0
channel        0
campaign       0
impressions    0
clicks         0
spend          0
conversions    0
revenue        0
dtype: int64
```

```
print("Duplicate Values:")
print(df.duplicated().sum())
```

```
Duplicate Values:
0
```

```
print("Statistics Summary")
df.describe()
```

Statistics Summary

| | date | impressions | clicks | spend | conversions | revenue |
|---|---|---|---|---|---|---|
| **count** | 260 | 260.000000 | 260.000000 | 260.000000 | 260.000000 | 260.000000 |
| **mean** | 2026-03-05 15:02:46.153846272 | 44709.826923 | 1516.030769 | 2816.422615 | 97.776923 | 12031.902308 |
| **min** | 2026-01-01 00:00:00 | 662.000000 | 11.000000 | 34.870000 | 0.000000 | 0.000000 |
| **25%** | 2026-02-02 00:00:00 | 22817.000000 | 507.250000 | 730.530000 | 30.000000 | 2749.485000 |
| **50%** | 2026-03-06 00:00:00 | 45528.000000 | 1253.500000 | 1898.140000 | 66.000000 | 6815.515000 |
| **75%** | 2026-04-06 06:00:00 | 66554.000000 | 2255.250000 | 3714.100000 | 140.250000 | 17514.715000 |
| **max** | 2026-05-01 00:00:00 | 89788.000000 | 5020.000000 | 14614.290000 | 552.000000 | 69634.760000 |
| **std** | NaN | 25920.630826 | 1163.146156 | 2915.208276 | 93.135411 | 13576.574893 |

```
print(df['channel'].unique())
```

```
['Organic' 'Email' 'Meta Ads' 'TikTok' 'Google Ads' 'Affiliate' 'LinkedIn']
```

```
df.groupby("channel")['clicks'].sum()
```

| channel | clicks |
|---|---|
| **Affiliate** | 41980 |
| **Email** | 66649 |
| **Google Ads** | 50683 |
| **LinkedIn** | 77931 |
| **Meta Ads** | 36388 |
| **Organic** | 65060 |
| **TikTok** | 55477 |

**dtype:** int64

```
df.groupby("channel")['revenue'].sum()
```

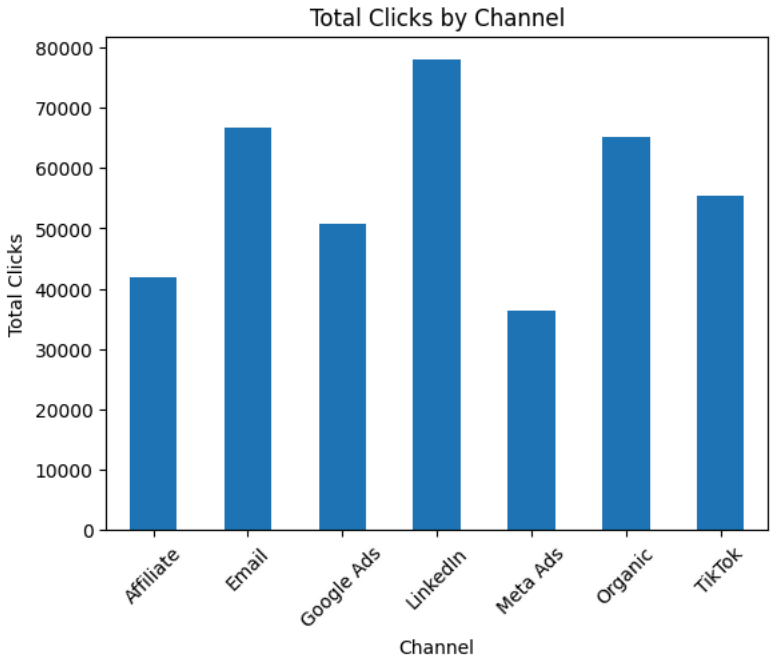
| channel | revenue |
|---|---|
| **Affiliate** | 425789.82 |
| **Email** | 509426.90 |
| **Google Ads** | 350750.79 |
| **LinkedIn** | 665081.15 |
| **Meta Ads** | 412845.02 |
| **Organic** | 426282.41 |
| **TikTok** | 338118.51 |

**dtype:** float64

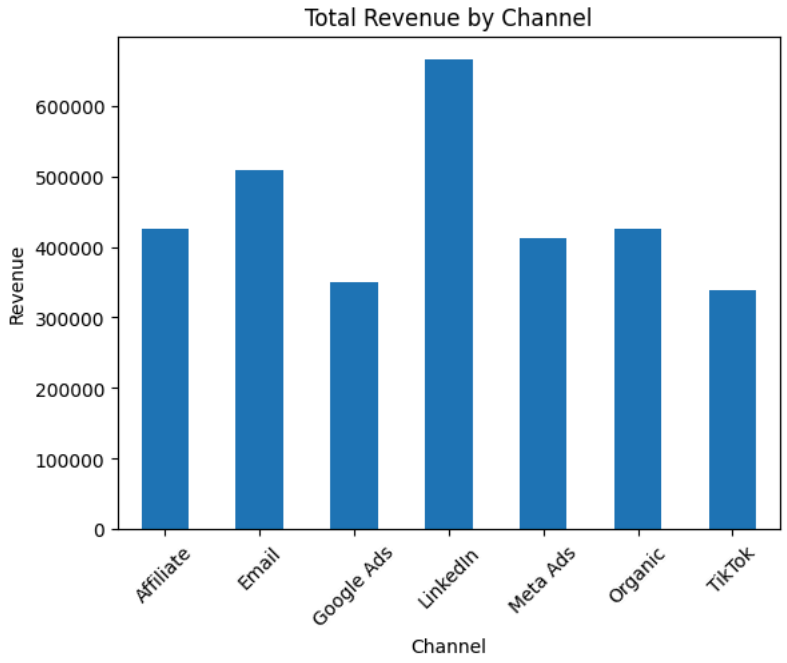
```
df.groupby("channel")[["clicks", "conversions", "spend", "revenue"]].sum()
```

| channel | clicks | conversions | spend | revenue |
|---|---|---|---|---|
| **Affiliate** | 41980 | 3091 | 90296.74 | 425789.82 |
| **Email** | 66649 | 3896 | 139415.51 | 509426.90 |
| **Google Ads** | 50683 | 3498 | 98350.33 | 350750.79 |
| **LinkedIn** | 77931 | 4832 | 128423.06 | 665081.15 |
| **Meta Ads** | 36388 | 3098 | 65394.72 | 412845.02 |
| **Organic** | 65060 | 4169 | 111037.12 | 426282.41 |
| **TikTok** | 55477 | 2838 | 99352.40 | 338118.51 |

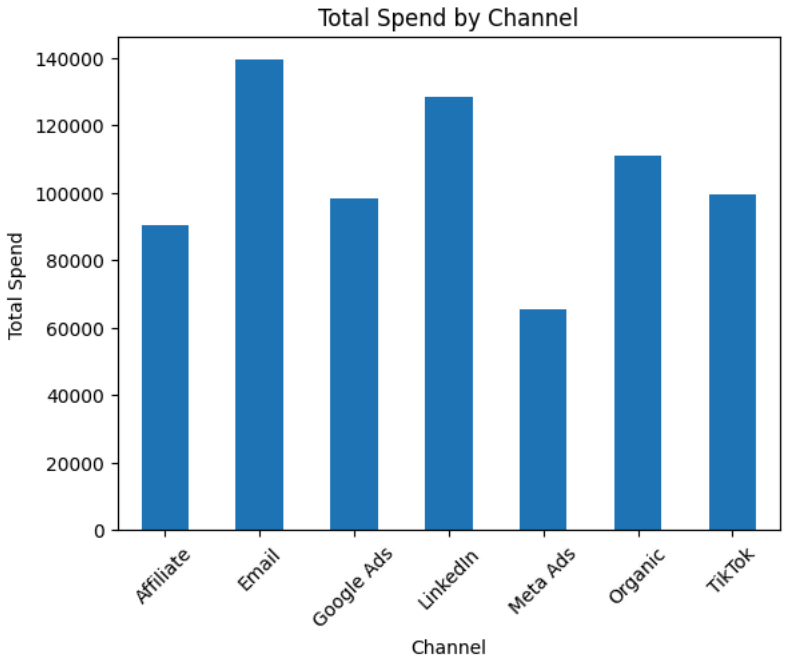
```
click=df.groupby("channel")["clicks"].sum()
click.plot(kind="bar", title="Total Clicks by Channel")
plt.xlabel("Channel")
plt.ylabel("Total Clicks")
plt.xticks(rotation=45)
plt.show()
```



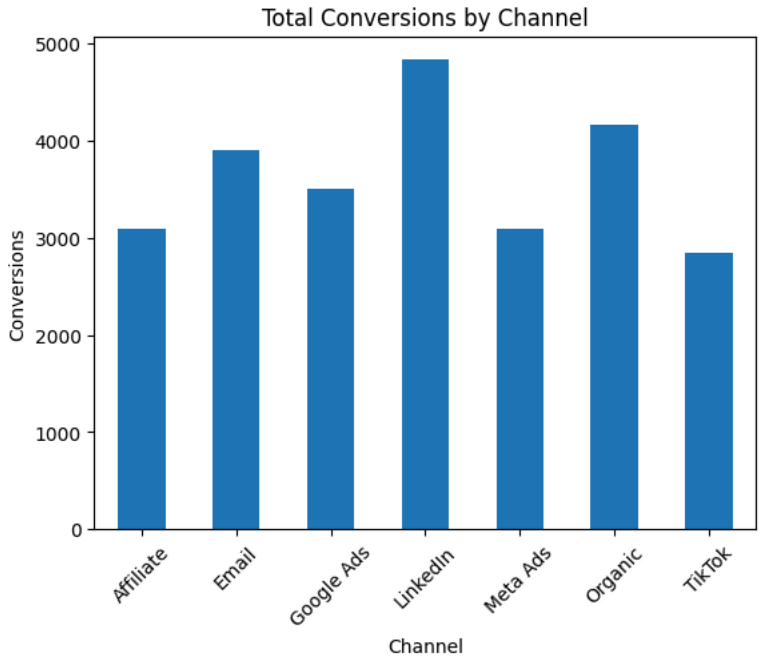

```
revenue=df.groupby("channel")["revenue"].sum()
revenue.plot(kind="bar", title="Total Revenue by Channel")
plt.xlabel("Channel")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.show()
```




```
spend=df.groupby("channel")["spend"].sum()
spend.plot(kind="bar", title="Total Spend by Channel")
plt.xlabel("Channel")
plt.ylabel("Total Spend")
plt.xticks(rotation=45)
plt.show()
```




```
conversions=df.groupby("channel")["conversions"].sum()
conversions.plot(kind="bar", title="Total Conversions by Channel")
plt.xlabel("Channel")
plt.ylabel("Conversions")
plt.xticks(rotation=45)
plt.show()
```




```
#Performance Metrics
total_clicks=df["clicks"].sum()
total_conversions=df["conversions"].sum()
total_spend=df["spend"].sum()
total_revenue=df["revenue"].sum()
print("Total Clicks:",total_clicks)
print("Total Conversions:",total_conversions)
print("Total Spend:",total_spend)
print("Total Revenue:",total_revenue)

if total_clicks!= 0:
    print("Conversion Rate (%):",total_conversions/total_clicks*100)
```

```
if total_conversions !=0:
    print("Cost per Conversion:",total_spend/total_conversions)

if total_spend != 0:
    print("ROAS:",total_revenue/total_spend)
```

```
Total Clicks: 394168
Total Conversions: 25422
Total Spend: 732269.88
Total Revenue: 3128294.5999999996
Conversion Rate (%): 6.449534208763776
Cost per Conversion: 28.80457399103139
ROAS: 4.272051446387498
```

```
channel_analysis=df.groupby("channel").agg({
    "clicks":"sum",
    "conversions":"sum",
    "spend":"sum",
    "revenue":"sum"
})

channel_analysis["conversion_rate"]=(
    channel_analysis["conversions"]/channel_analysis["clicks"]*100
)

channel_analysis["cost_per_conversion"]=(
    channel_analysis["spend"]/channel_analysis["conversions"].replace(0, np.nan)
)

channel_analysis["ROAS"]=(
    channel_analysis["revenue"]/
    channel_analysis["spend"].replace(0,np.nan)
)
channel_analysis.round(2)
```

| channel | clicks | conversions | spend | revenue | conversion_rate | cost_per_conversion | ROAS |
|---|---|---|---|---|---|---|---|
| **Affiliate** | 41980 | 3091 | 90296.74 | 425789.82 | 7.36 | 29.21 | 4.72 |
| **Email** | 66649 | 3896 | 139415.51 | 509426.90 | 5.85 | 35.78 | 3.65 |
| **Google Ads** | 50683 | 3408 | 98250.33 | 350750.70 | 6.90 | 28.13 | 3.57 |